T02 - Heat Transfer and Thermal Calculations

REGIONAL WEBSITE RELEASE 1.0

Name/team: _____ Date: _____

24 questions | 60 points

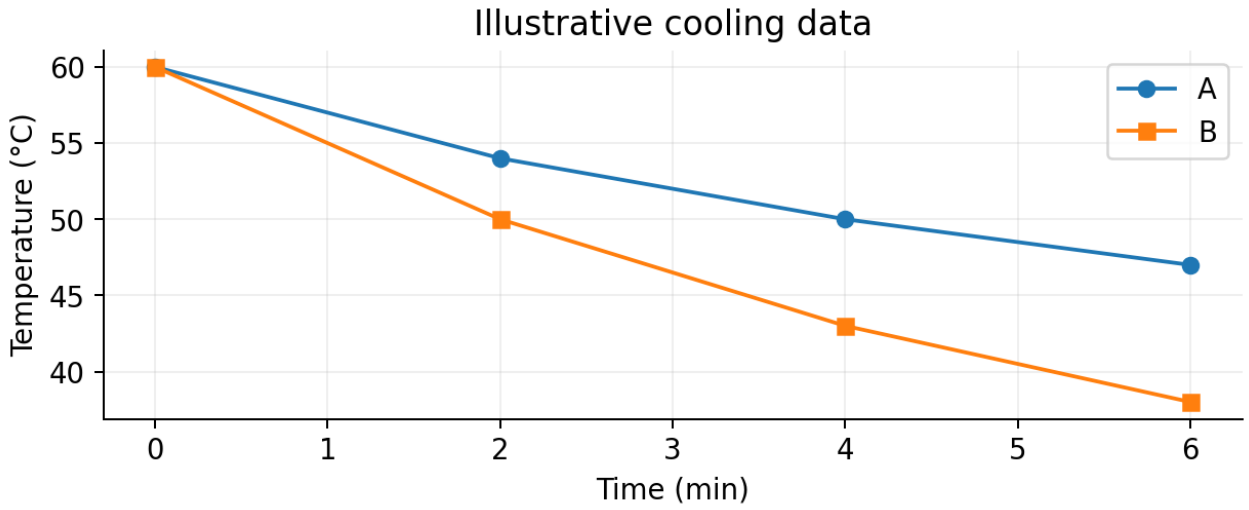
Use metric units and show reasoning. Suggested practice time: 45–55 minutes. These are instructional practice timings, not official event allocations. The full official event includes device testing. Use 273.15 for Celsius-to-Kelvin conversion and supplied constants. Q is positive into the system; W is positive by it; $\Delta U = Q - W$. Closed-system energy calculations neglect bulk kinetic/potential changes unless stated. No calculus, trigonometry, logarithms or State/National-only radiation/third-law calculations are required. All datasets are original illustrative models.

Tests 1–3 are focused practice and are not full official written-test simulations. Test 4 covers all five Regional written categories; device items use supplied foundational models and do not require deferred Unit 7.

Reference pages precede the question pages. Each question page has space for two responses. Use extra paper if needed and label every answer with its question ID. Answers and grading rubrics are in the separate instructor edition.

T02-F1 - Cooling curves

Recorded-style teaching data for two samples cooling toward a room colder than both. No historical or official device measurements are claimed.



Time (min)	A (°C)	B (°C)
0	60	60
2	54	50
4	50	43
6	47	38

T02-D2 – Steady slab data

Same material, cross-sectional area and face temperature difference for each row; steady one-dimensional model.

Thickness (m)	Transfer rate (W)
0.01	36
0.02	18
0.04	9

T02 – Questions 1–2

Name/team: _____

Show reasoning, equations and units. Use the reference pages as needed.

T02-Q01 (2 points) A metal spoon handle warms when its other end rests in warmer water. What mechanism carries energy along the stationary spoon? Choose and explain.

A. Natural convection

B. Conduction

C. Evaporation only

D. Work by a piston

T02-Q02 (2 points) Which example is forced convection? Choose and explain.

A. Energy through a solid rod

B. Radiation through vacuum

C. Fan-driven airflow

D. Buoyant warm air rising without a fan

T02 - Questions 3–4

Name/team: _____

Show reasoning, equations and units. Use the reference pages as needed.

T02-Q03 (2 points) A warmer object transfers energy to a cooler object across a vacuum gap. Which mechanism can do this? Choose and explain.

A. Radiation

B. Convection

C. Conduction through air

D. Liquid circulation

T02-Q04 (2 points) Which unit belongs to thermal conductivity k? Choose and explain the quantity it describes.

A. J

B. J/K

C. J/(kg·K)

D. W/(m·K)

T02 - Questions 5–6

Name/team: _____

Show reasoning, equations and units. Use the reference pages as needed.

T02-Q05 (2 points) A body has heat capacity 360 J/K. Neglect losses and work. How much heat raises its temperature by 4.5 K?

T02-Q06 (2 points) A 0.300 kg sample has c=750 J/(kg·K). Calculate its heat capacity.

T02 - Questions 7–8

Name/team: _____

Show reasoning, equations and units. Use the reference pages as needed.

T02-Q07 (2 points) Explain why natural convection can occur in a liquid but does not describe energy transfer through a solid block that remains fixed in shape.

T02-Q08 (2 points) A polished metal surface reflects much of the incident thermal radiation in a specified wavelength range. Does this mean it creates energy? Explain the energy accounting for an opaque surface.

T02 – Questions 9–10

Name/team: _____

Show reasoning, equations and units. Use the reference pages as needed.

T02-Q09 (2 points) Use the T02 cooling graph. Which sample cools more rapidly during the first 2 minutes, and what are the two average temperature-change rates?

T02-Q10 (2 points) Equal masses of two materials receive equal net energy without a phase change. Sample X warms less than Y. Which has the larger specific heat, and why?

T02 – Questions 11–12

Name/team: _____

Show reasoning, equations and units. Use the reference pages as needed.

T02-Q11 (2 points) Name the dominant transfer mechanism for each described segment: a lamp sends energy across air to a plate; energy moves through the solid plate; heated water circulates above the plate.

T02-Q12 (2 points) A 15 W heater operates for 80 s. What energy does it supply? State why 15 W is not an energy amount.

T02 – Questions 13–14

Name/team: _____

Show reasoning, equations and units. Use the reference pages as needed.

T02-Q13 (3 points) A plane slab has k=0.060 W/(m·K), A=0.50 m², thickness L=0.030 m and a steady temperature difference of 18 K. Use P=kAΔT/L to find its conductive transfer rate.

T02-Q14 (3 points) For the slab in Q13, find energy transferred in 5.0 minutes if the rate remains constant. If thickness doubles with all other model quantities unchanged, what is the new rate?

T02 – Questions 15–16

Name/team: _____

Show reasoning, equations and units. Use the reference pages as needed.

T02-Q15 (3 points) A 0.180 kg sample of water warms from 21.0 °C to 29.0 °C. With c=4200 J/(kg·K), find its thermal energy gain in the negligible-work model.

T02-Q16 (3 points) A body absorbs 2340 J and warms by 13.0 K. Its mass is 0.400 kg. Neglect losses and phase changes. Find c and state its unit.

T02 – Questions 17–18

Name/team: _____

Show reasoning, equations and units. Use the reference pages as needed.

T02-Q17 (3 points) A small water sample and its cup have heat capacities 420 J/K and 60 J/K respectively. They warm together by 7.0 K. Neglect losses. Find their total energy gain and the error from counting water alone.

T02-Q18 (3 points) Explain how trapped air can help an insulating layer, and why a large open air space that circulates freely may perform differently.

T02 – Questions 19–20

Name/team: _____

Show reasoning, equations and units. Use the reference pages as needed.

T02-Q19 (3 points) Two surfaces have equal geometry and temperature and receive identical incident radiation. In the relevant spectral range A absorbs a larger fraction than B. Can you conclude A always reaches a higher final temperature in every setup? Explain.

T02-Q20 (3 points) In the T02 graph, sample A has mass 0.100 kg and c=4200 J/(kg·K). Find the magnitude of its energy loss over the first 4 minutes and its average outward power. Neglect evaporation and cup energy.

T02 – Questions 21–22

Name/team: _____

Show reasoning, equations and units. Use the reference pages as needed.

T02-Q21 (3 points) A team compares two insulating covers, but changes water mass, initial temperature and airflow at the same time. Identify two controls and explain why the comparison cannot isolate the cover’s effect.

T02-Q22 (3 points) A warm body loses 7 W by convection and 5 W by radiation while receiving 20 W from a heater. Other transfers are negligible and rates are constant for 90 s. Find its net energy gain.

T02 – Questions 23–24

Name/team: _____

Show reasoning, equations and units. Use the reference pages as needed.

T02-Q23 (3 points) In two steady slab tests, area and temperature difference are fixed. The T02 table lists thickness 0.010, 0.020 and 0.040 m with powers 36, 18 and 9 W. Does P×L stay constant? Predict the power at 0.030 m under this model and state one assumption.

T02-Q24 (3 points) A student says, “The material with the greatest specific heat must also conduct fastest.” Explain why the claim does not follow, including the units of both quantities and the different experiments needed.